ESYS 104: The Earth System
Assignment #2

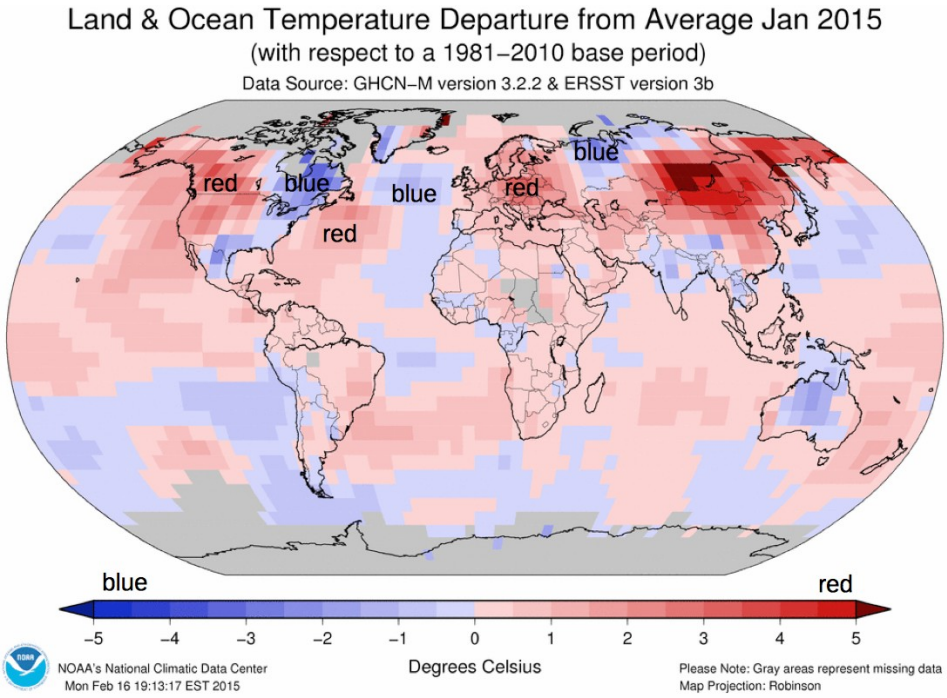
Due **October 19, 2026**. Please don't submit photos of handwritten text answers -use a text editor (photos of drawings are OK)(Legibility is important! If you handwriting is very very good, then that's OK). Late assignments will lose 10% for each day/fraction of a day it's late. You may work together, but **your answers must be unique and written in your own words**.

Short Answer (two-three sentences, max) for all questions.

Question 1: Weather
In 2015, the eastern US and Canada was much colder than normal. Considering the figure below, which shows the deviation from averages temperatures (hotter or colder than normal):

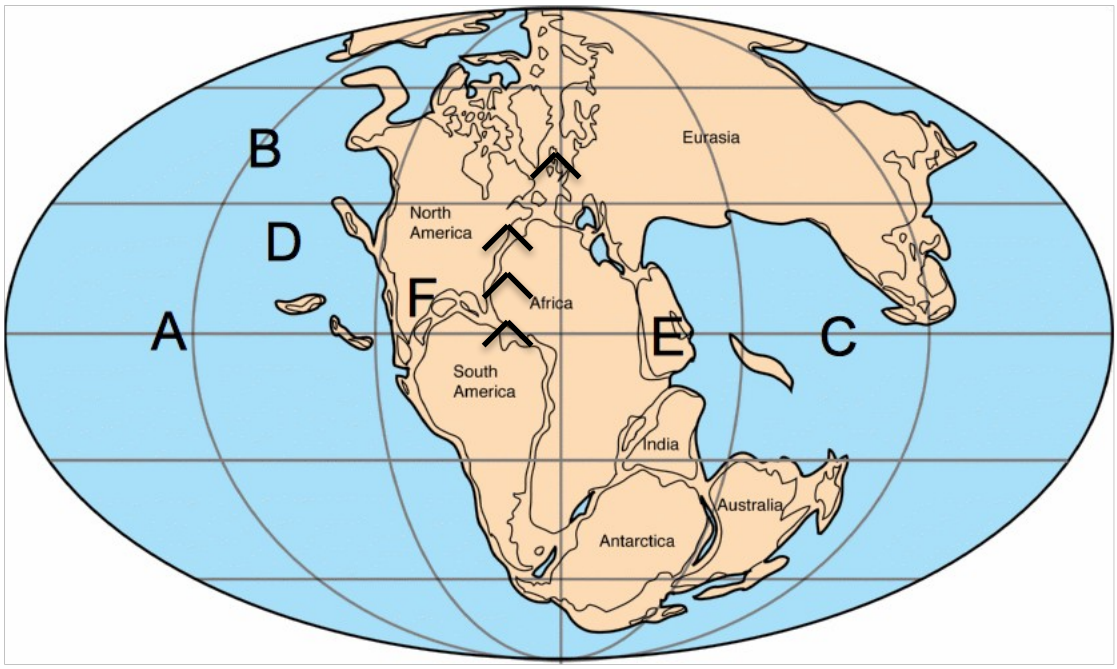
a) Draw or indicate the Jan-Feb mean Jet Stream location over North America and the Atlantic on the map below with a thick line.

b) Would you say that the Polar Vortex (circulation in the polar cell of the three-cell model) over North America in January and February of 2015 was weak or strong? Explain.



Question 2: Climate

Continental drift is currently making the Atlantic wider; if you go back in time, the Atlantic Ocean disappears, and all the continents were joined together. Below is a reconstruction of the Supercontinent Pangea, during the Age of Dinosaurs. The horizontal grey lines indicate the equator, ±30° and ±60° latitude. The **high** mountains (^) of the young Appalachians lay between Africa and North America.



a) for the locations marked with the letters 'A', 'B', & 'C' indicate the dominant wind direction, based on your knowledge of Hadley cells.

b) at location 'D', what is the likely surface ocean current direction?

c) In terms of temperature and precipitation, what is the climate at location 'E'?

d) What is the climate at location 'F'

e) Was Antarctica covered in ice during this time? Describe or sketch a possible ocean current that would transport equatorial heat towards Antarctica.

Question 3-5: watch the following videos (from 'Minute Earth') and answer the question (1-3 sentences, max):

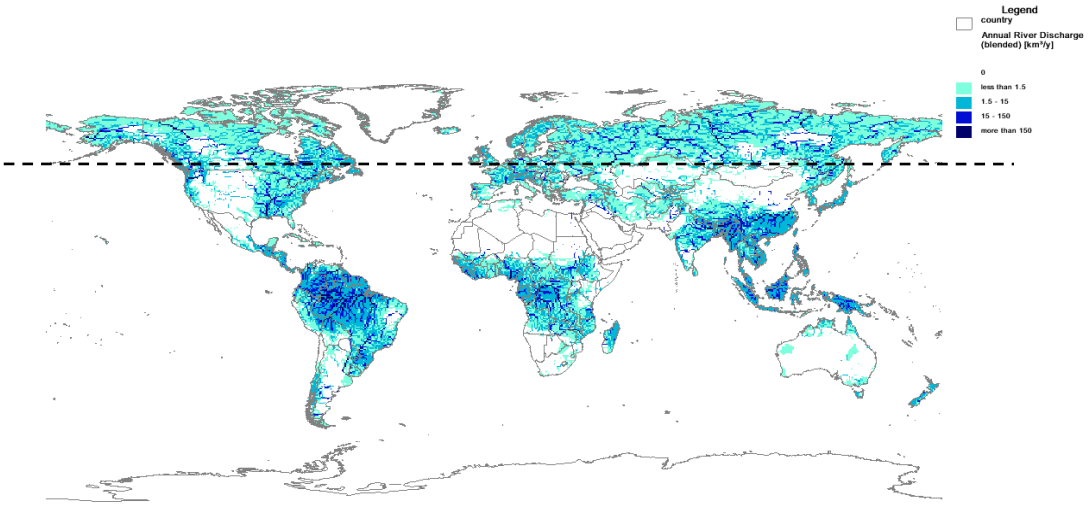
Why Do Rivers Curve?
https://www.youtube.com/watch?v=8a3r-cG8Wic

Why Do Rivers Have Deltas?
https://www.youtube.com/watch?v=A47ythEcz74

Which Came First - The Rain or the Rainforest?
https://www.youtube.com/watch?v=Y3OWgb0Bv-A

Question 6: Fresh Water

Below is a global map of river **discharge** (river flow volume). Darker indicates more average annual flow.



a) Why is river discharge at high latitudes (say northern Canada, above the dashed line) higher than at mid-latitudes (i.e. Canadian Prairies, below the line) even though the Prairies get more precipitation?

b) Why does river discharge vary within latitude bands (i.e. more or less along the same latitude)? What can cause those differences?

Question 7: Biomes and Food chains

a) In the Biomes diagram below, why is there no biome identified for 0°C mean annual temperature and 300 cm of annual precipitation? (In 1-2 sentences maximum)

